- There are 7 Questions for 50 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof). Clearly justify any statement you make, even if the problem doesn't explicitly ask you to.
- Good luck!

1. (a) [2] Given $m \times n$ matrices A, B and $\underline{y} \in \mathbb{R}^n$, what is the vector $\underline{x} \in \mathbb{R}^n$ that minimizes $\|A\underline{x} - B\underline{y}\|_2$?
   (b) [2] Suppose that any column of A is orthogonal to any column of B. How would you give a 'simple' equation (involving matrices A, B, transposes and matrix products) that captures this relationship ?
   (c) [2] In case A, B satisfy the property in (b) above, what is the corresponding answer to (a) ?

2. Consider the following sequence of matrices

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_n = \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix} \text{ for } n \geq 2.$$

   (a) [2] What is the size of the matrix A_n ?
   (b) [2] Is A_n symmetric ?
   (c) [2] What is A_n^2 ? *Hint: Guess a pattern, then prove by induction*
   (d) [2] What is the rank of A_n ?

3. Consider the $(n+1) \times (n+1)$ matrix

$$A = \begin{pmatrix} I & \underline{a} \\ \underline{a}^\mathsf{T} & 0 \end{pmatrix},$$

   where $\underline{a} \in \mathbb{R}^n$ and I is $n \times n$ Identity matrix.
   (a) [3] When is A invertible? Give your answer in terms of $\underline{a}$. Justify your answer.
   (b) [3] Assuming the condition you found above holds, give an expression for the inverse matrix. *Hint:* What is the answer when $n = 1$?

4. For this problem, A is a 2×2 matrix. Answer true or false, with justification. All of the parts are separate problems. If the statement is false, provide a counterexample. If true, provide proof of the same.
   (a) [2] If $A^2 = 0$ then A cannot be symmetric nonzero.
   (b) [2] If A has diagonal entries 1, and $\underline{x}$ is a unit norm vector ($\|x\|_2 = 1$) then we can pick $\underline{x}$ such that $\underline{x}^T A \underline{x} = 1$.
   (c) [2] Similar to above, if A has diagonal entries 1, and $\underline{x}$ is a unit norm vector ($\|x\|_2 = 1$) then we cannot pick $\underline{x}$ such that $\underline{x}^T A \underline{x} > 1$.

5. Let A be an $m \times n$ matrix. Let k an integer such that A can be factored as

$$A = BC, \quad \text{where } B \text{ is } m \times k \text{ and } C \text{ is } k \times n. \tag{1}$$

   (a) [3] Show that k has to be at least $\text{rank}(A)$, i.e. $k \geq \text{rank}(A)$.
   (b) [3] When $k = \text{rank}(A)$, how would you compute the decomposition (1) above ? How many arithmetic operations would such a computation cost?

(c) [3] Suppose A is an $m \times n$ matrix with rank k, and a decomposition like in (1) has already been computed (i.e. B and C are known). Now, we would like to compute $A\underline{x}$ for inputs $\underline{x} \in \mathbb{R}^n$. How would you do this computation? How many arithmetic operations would it cost ?

6. [8] Consider a robot with unit mass that can move in a frictionless two-dimensional plane. The robot has a constant unit speed in the $y-$direction (towards north), and it is designed such that we can only apply force in the $x-$direction. We will apply a (horizontal) force at time t given by f_j for $2j-2 \leq t \leq 2j$ where $j = 1, ..., n$, so that the applied force is constant over time intervals of length 2. So a force of f_1 is applied for time $0 \leq t \leq 2$, force f_2 for time $2 \leq t \leq 4$ and so on.

   In this problem, we assume that $y_i = i$, the robot is initially at $(0,0)$, and the initial velocity of the robot has no $x-$component.

   (a) Find the coordinates of the robot at time t, where $0 \leq t \leq 2n$. Your answer should be a function of t and the vector of input forces $\underline{f} \in \mathbb{R}^n$. You do not need the exact expression, but you need to correctly identify the structure of the position function (with variables as the forces applied $\underline{f}$).

   Now, there are $2n$ coins in the plane and the goal is to design a sequence of input forces for the robot to collect the maximum possible number of coins. The robot is designed to collect the i^{th} coin only if it passes through the location of the coin (x_i, y_i).

   (b) Given the location of the 2n coins, describe a method to determine if it is possible for the robot to collect them.

   Hint: With initial position x_0, initial velocity v_0 and constant acceleration a, the position and velocity at time t are given by $x(t) = x_0 + v_0 t + \frac{1}{2}at^2, \quad v(t) = v_0 + at$.

7. [7] What is the nearest neighbor of $x \in \mathbb{R}^n$ among the unit vectors $e_1, ..., e_n$ in the $p-$ norm sense, for $p = 1, 2, \infty$?